Nalla Nandan Santosh

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.17
Sri Chaitanya Junior College	Class XII (Telangana State Board)	2024	98.20%
Sri Chaitanya High School	Class X (Telangana State Board)	2022	9.80%

SCHOLASTIC ACHIEVEMENTS

Department **Rank 6** in B.Tech Mathematics and Computing.

Qualified Stage 1 of **AlgoUniversity Tech Fellowship (ATF) 2025**; ranked in **top 8%** nationwide among **250,000+** participants and advanced to Stage 2.

Secured **Rank 51** in Optiver Trade-a-thon (Quantitative Trading Competition).

Qualified for **INChO (2nd level)** through **NSEC - National Chemistry Olympiad** pathway

PROJECTS

Alpha Factor Discovery and Backtesting Framework

- Developed an end-to-end quantitative research pipeline in **Python** using **Pandas, Scikit-Learn, and Yahoo Finance**, collecting and preprocessing five years of historical OHLCV data across **10 NSE-listed stocks** for machine learning-based return prediction.
- Engineered multiple technical alpha factors, including **momentum, volatility, RSI, moving-average ratios, and volume-based indicators**, and transformed raw market data into predictive features for forecasting future stock returns.
- Trained and evaluated **Linear Regression** and **Random Forest** models using a time-based out-of-sample train/test split, assessing predictive performance through **Mean Squared Error (MSE)**, correlation analysis, and model comparison.
- Constructed stock-ranking portfolios from model predictions and performed historical backtesting with an equal-weight benchmark, analyzing cumulative returns and portfolio performance to evaluate the effectiveness of the engineered factors.

Keyword Tree-based Pattern Matching (Aho-Corasick)

- Implemented the **Aho-Corasick algorithm** in C++ by constructing Keyword Trees (tries) and failure links to efficiently perform multi-pattern string matching.
- Developed a pattern-matching system for large DNA sequences, locating all occurrences of multiple substrings in **O(n + m + k)** time while correctly handling overlapping pattern matches.
- Evaluated the algorithm by comparing execution time and character-comparison counts against a naive shift-based string matching approach, demonstrating improved efficiency and scalability on large datasets.

Statistical Data Analysis and Inference

- Conducted exploratory data analysis on multiple real-world datasets by constructing frequency distributions, histograms, box plots, frequency polygons, and pie charts to summarize and visualize data characteristics.
- Applied statistical inference techniques including hypothesis testing, confidence intervals, chi-squared tests, and sampling-based analysis to evaluate population characteristics and validate statistical assumptions.
- Implemented **Maximum Likelihood Estimation (MLE)**, **Method of Moments (MOM)**, contingency table analysis, and linear regression models using **NumPy, Pandas, and Matplotlib** for parameter estimation and predictive analysis.

Hasse Diagram Generator for Partially Ordered Sets

- Developed a graph-based application in C++ to construct Hasse diagrams for partially ordered sets (posets), representing order relations by eliminating redundant transitive edges.
- Implemented probability propagation and aggregation algorithms across posets using subset-superset relationships, enabling efficient computation of cumulative probabilities over hierarchical structures.
- Applied concepts from discrete mathematics, graph theory, and order theory to analyze structured datasets, validate partial-order properties, and visualize hierarchical relationships effectively.

SKILLS

Programming Languages: C, C++, Python, SQL, HTML, CSS

Tools and Libraries: STL, Pandas, NumPy, MATLAB, LaTeX

Competitive Programming: Codeforces (Rating 1310, 200+ Problems Solved)

Soft Skills : Team Work, Leadership, Communication and Negotiation.

RELEVANT COURSES

Computer Science: Introduction to Programming, Data Structures, Algorithms, Theory of Computation, DBMS-I, Compilers-I

Applied Mathematics : Linear Algebra, Probability, Applied Statistics, Convex Optimization

Pure Mathematics : Math Foundation, Calculus, Number Theory, Differential Equations, Transform Techniques, Groups & Rings , Metric Spaces.

Electrical Engineering : Linear Systems and Signal Processing, Basic Electrical Engineering

POSITIONS OF RESPONSIBILITIES

OCS Internship Coordinator

Class Representative - Mathematics and Computing Branch

EXTRACURRICULAR AND HOBBIES

NSO - Lawn Tennis Member

Chess (Max Rating - 1236, Chess.com), Mathematical Puzzle Solving

Tennis (8+ years of playing experience), Badminton